
Exercises: Advanced Lesson 13

Your Name: Mikey Ferguson

Your Math Class: MATH 7013 - Advanced Mathematical Modeling

1: Use the Trace command to analyze the intermediate steps in the following commands:

$$\begin{aligned} &\partial_x x^2 \\ &\partial_x (1+x)^2 \\ &\partial_x (x+1)^2 \\ &\int (x+x^2+e^x) dx \\ &\partial_x \left(\int (1+x)^2 dx \right) \end{aligned}$$

Clear[x];

Trace[$\partial_x x^2$]

Trace[$\partial_x (1+x)^2$]

Trace[$\partial_x (x+1)^2$]

Trace[$\int (x+x^2+e^x) dx$]

Trace[$\partial_x \int (1+x)^2 dx$]

2: Use the Trace command to analyze the intermediate steps in the following commands:

Part 1:

Clear[a,b]

Trace[Sin[a] /. a → b]

Part 2:

Clear[a,b]

b=5

Trace[Sin[a] /. a → b]

Part 3:

Clear[a,b]

a=2

b=5

Trace[Sin[a] /. a → b]

Part 4:

Clear[a,b]

Trace[Sin[a] /. a → b]

Part 5:

```
Clear[a,b]
b=5
Trace[ Sin[a] /. a -> b]
```

Part 6:

```
Clear[a,b]
a=2
b=5
Trace[ Sin[a] /. a -> b]
```

Observe that the lhs of $a \rightarrow b$ is not held, however replacing "a" in both the expression $\text{Sin}[a]$ and the lhs of the delayed replacement rule has no effect upon the final answer.

```
Clear[a, b];
Trace[Sin[a] /. a -> b]
```

```
Clear[a, b];
b = 5;
Trace[Sin[a] /. a -> b]
```

```
Clear[a, b];
a = 2;
b = 5;
Trace[Sin[a] /. a -> b]
```

```
Clear[a, b];
Trace[Sin[a] /. a -> b]
```

```
Clear[a, b];
b = 5;
Trace[Sin[a] /. a -> b]
```

```
Clear[a, b];
a = 2;
b = 5;
Trace[Sin[a] /. a -> b]
```

3: Construct a function $f[x_1, x_2, x_3, \dots, x_n]$ which will accept a Sequence of expressions:

$x_1, x_2, x_3, \dots, x_n$ (i.e. not a list) and display a list of a list of both the unevaluated and evaluated form of each element x_i , $i = 1, \dots, n$.

Test your function:

$$f[N[\pi], 1 + 2, a * a^2, a * (1 + a)]$$

$$\{\{N[\pi], 3.14159\}, \{1 + 2, 3\}, \{a a^2, a^3\}, \{a (1 + a), a (1 + a)\}\}$$

and:

```
f[N[ $\pi$ ], 1 + 2,  $a * a^2$ ,  $a * (1 + a)$ ] // TableForm
```

```
N[ $\pi$ ]      3.14159
1 + 2      3
a a2      a3
a (1 + a)  a (1 + a)
```

(No hints will be given for this problem.)

```
ClearAll[f, a];
SetAttributes[f, HoldAll];
f[args___] := List @@ Function[u, {HoldForm[u], u}, HoldAll] /@ Hold[args]

Clear[a];
f[N[ $\pi$ ], 1 + 2, a a2, a (1 + a)]
TableForm[f[N[ $\pi$ ], 1 + 2, a a2, a (1 + a)]]
```